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AI for Bharat Hackathon

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Team Name : **Trinaari AI**

Team Leader Name : **Sowmya Akula**

Problem Statement : **KrishiMitra :"Empowering farmers with voice-powered intelligence"**

Problem Statement

Small and marginal farmers in India often suffer severe crop losses due to delayed detection of drought, pest infestation, disease, or irrigation failures. Existing systems are reactive, generic, or inaccessible due to language and literacy barriers.

There is no intelligent, multilingual, voice-first AI system that:

- **Monitors** crops **autonomously**
- **Predicts** stress before damage
- **Explains** problems clearly
- **Engages** farmers conversationally
- **Continuously learns** and **improves**

🌾 KrishiMitra

A Proactive, Agentic, Voice-First AI System for farmers

- Monitors crops using satellite + weather data,
- predicts stress 3+ days early, and
- calls farmers in their local language with personalized advice.

What Makes It Different?

⚡ AI-based early stress prediction
🧠 Agentic AI reasoning (not rule-based alerts)
🎙️ Multilingual voice chatbot
🔄 Continuous self-learning
🎯 False-alert reduction via autonomous investigation

Outcome

A self-learning agricultural intelligence network — not just an alert system.

Why AI is required ?

Agriculture problems are complex and require:

- Image understanding (crop disease detection)
- Natural language understanding
- Data-driven insights from weather and market data

AI enables:

- Automatic crop disease detection
- Intelligent farmer query answering
- Context-aware farming recommendations.

Value of AI Layer to User Experience

- Instant diagnosis instead of waiting for agricultural experts
- Personalized farming guidance
- Data-driven recommendations.

How AWS services are used ?

AWS provides scalable infrastructure for the solution:

AI & Intelligence Layer:
Amazon Bedrock
Powers AI advisory generation using Amazon Nova Lite LLM

Compute & Hosting:
AWS Elastic Beanstalk
Hosts FastAPI backend application (Python 3.11)

Amazon S3:
Hosts React frontend + stores audio files for voice calls

Data Storage:
SQLite on Elastic Beanstalk
Stores farmer profiles, farm data, and advisories

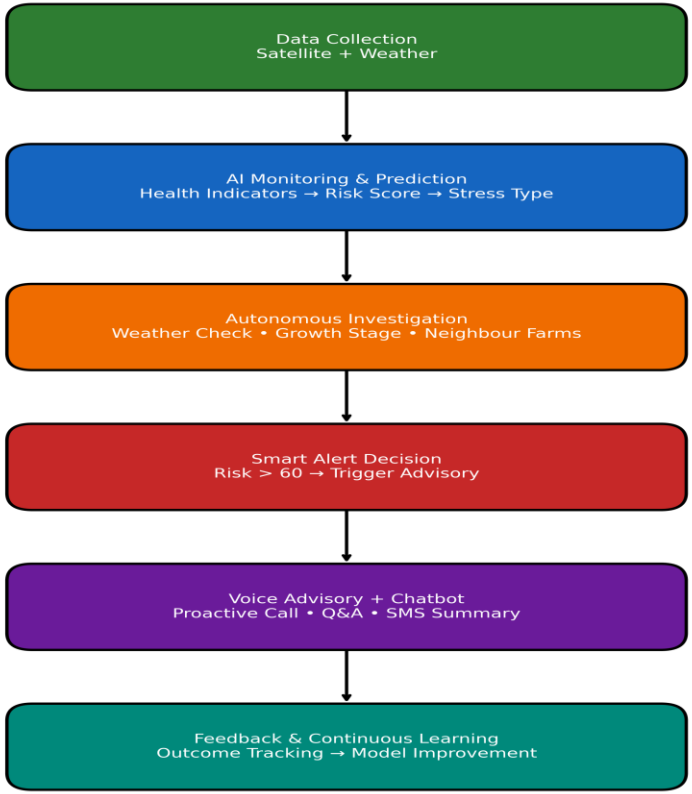
Security & Access:
AWS IAM
Secure access management for AWS services

Monitoring:
Amazon CloudWatch
Application monitoring and logging

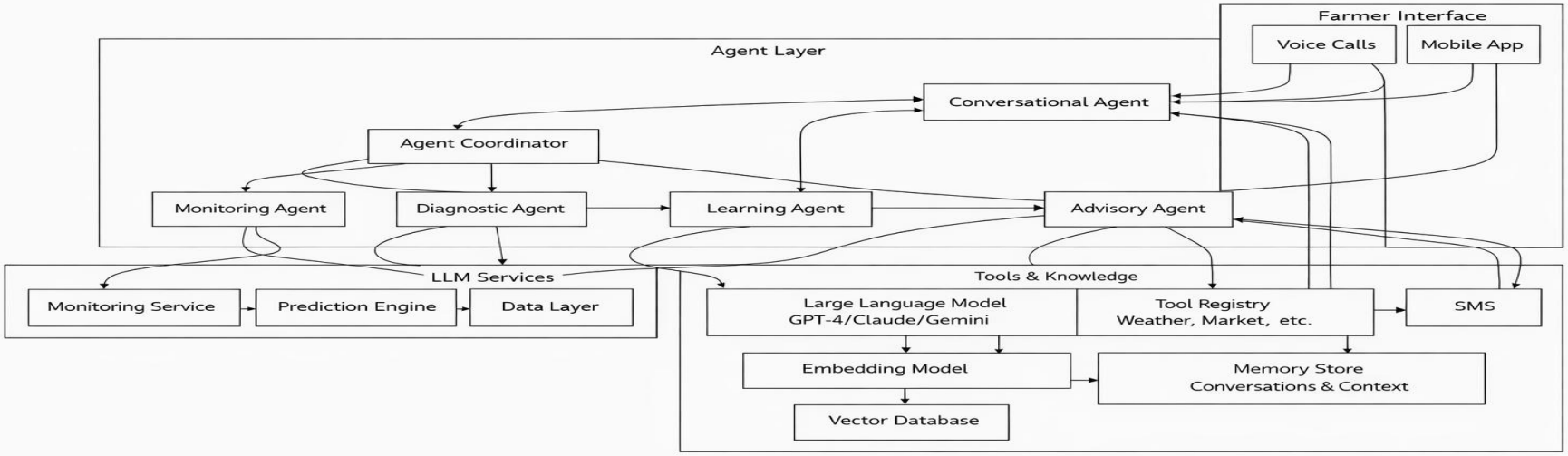
KrishiMitra – Key Features

- **Satellite & Weather Auto Monitoring**
  Continuous remote sensing every 3 days + 6-hour weather tracking
- **3-Day Early Stress Prediction**
  AI risk scoring with drought, pest, disease detection
- **Personalized Agentic AI Advisories**
  Budget-aware, resource-specific, explainable recommendations
- **Multilingual Proactive Voice Alerts**
  Auto-calls within 30 mins in 10+ Indian languages
- **24/7 Conversational Voice Chatbot**
  Natural, multi-turn support with <2 sec response time
- **Outcome Tracking & Smart Analytics**
  Call success, farmer feedback, regional impact insights
- **Self-Learning & False Alert Reduction**
  Autonomous investigation + continuous model improvement
- **Secure, Scalable, Cloud-Ready Architecture**
  Encrypted, resilient, supports 10,000+ farmers

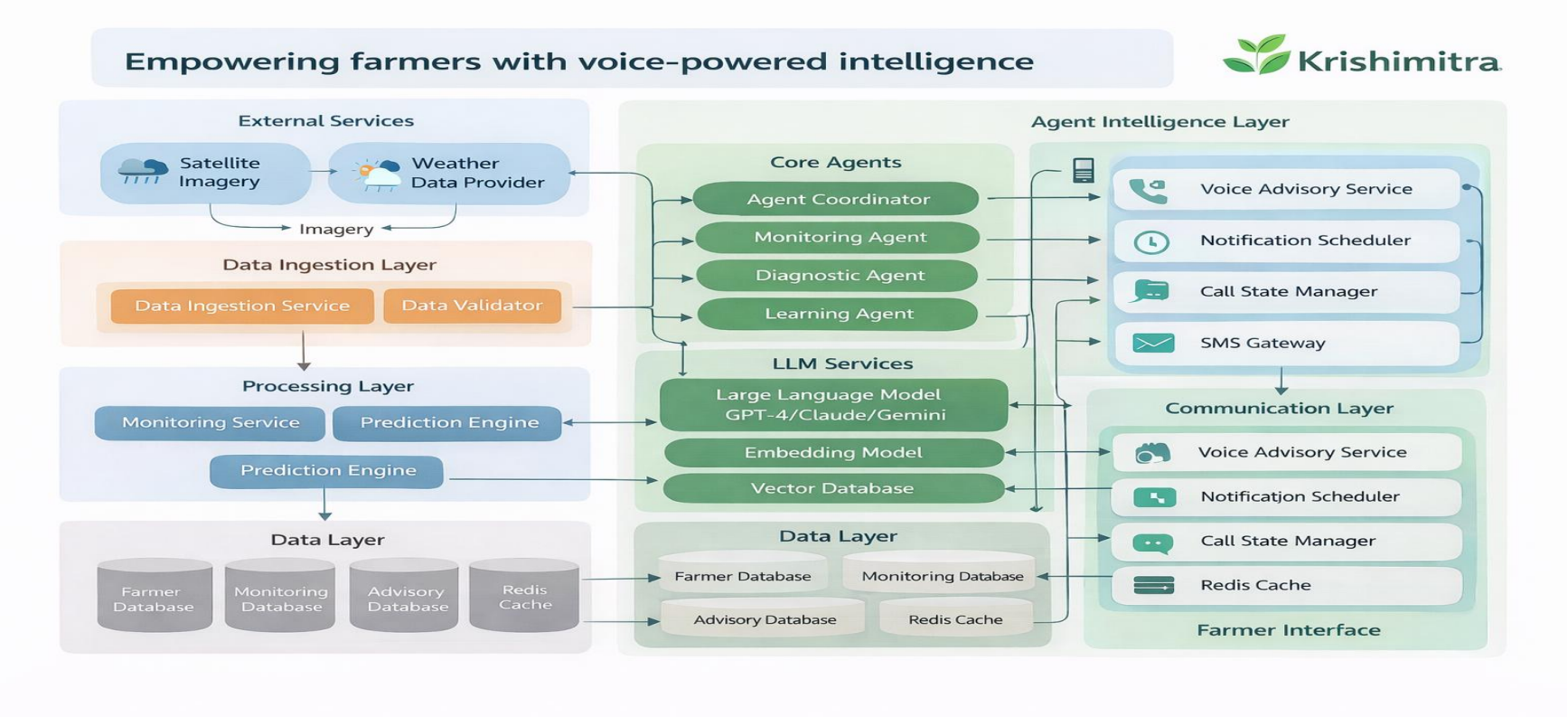
Process flow diagram



Flow diagram in Detail



Architecture diagram of the proposed solution:



AWS Architecture Overview

DATA INGESTION C

- Amazon API Gateway
- AWS Lambda (Scheduled Polling)
- Amazon S3 (Satellite + Weather Data Lake)

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AI & INTELLIGENCE LAYER

- Amazon SageMaker (Stress Prediction Model)
- Amazon Bedrock (Agentic AI Advisory + RAG)
- Amazon OpenSearch (Knowledge Base Search)

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VOICE & COMMUNICATION

- Amazon Connect (Outbound Voice Calls)
- Amazon Lex (Conversational Chatbot)
- Amazon Polly (Text-to-Speech)
- Amazon SNS (SMS Fallback Alerts)

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DATA STORAGE

- RDS (Farmer DB) • DynamoDB (Sessions)
- ElastiCache (Active Alerts)

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MONITORING & SECURITY

CloudWatch • IAM • KMS

Technologies utilized in the solution:

Frontend

- React

Backend

- Python
- FastAPI

AI & ML services

- LLM: AWS Bedrock (Amazon Nova Lite)
- Satellite Imagery: Sentinel Hub API
- NDVI Calculation: Custom implementation for crop health analysis

Cloud Infrastructure

- AWS Elastic Beanstalk (backend hosting)
- AWS S3 (frontend hosting + audio storage)
- AWS Bedrock (AI/LLM services)
- AWS IAM (access management)

External APIs

- OpenWeather API (weather data)
- Twilio (voice call delivery)
- Google TTS (text-to-speec)

Estimated Implementation Cost

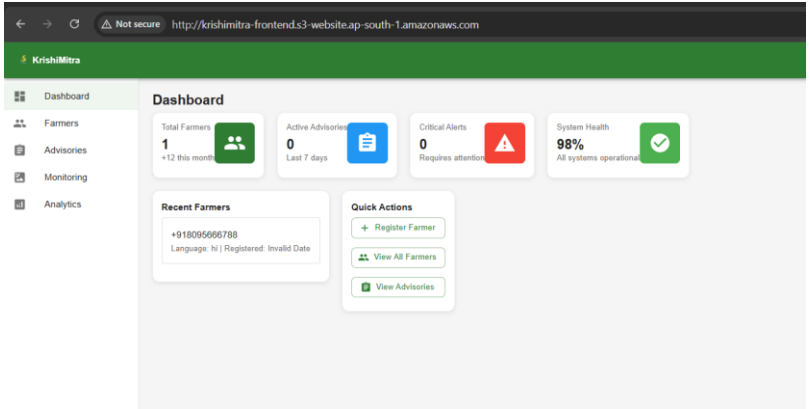
For 10,000 farmers (monthly estimate):

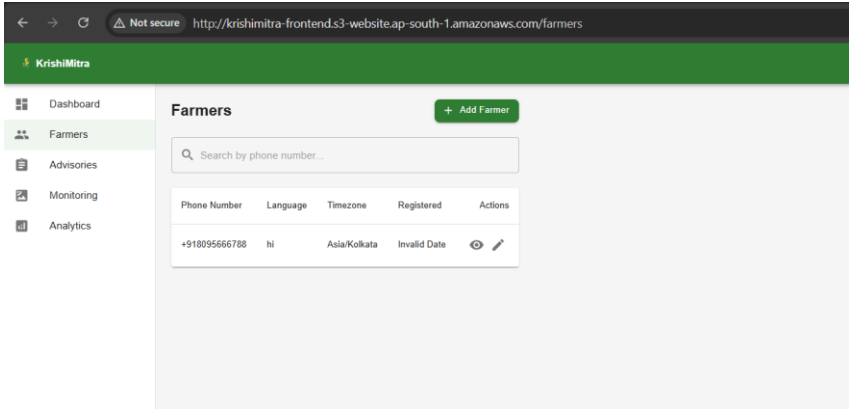
Component	Equivalent in INR (@ ₹83/$)
Sentinel Hub API	₹8,300
OpenWeather API	₹9,960
Twilio Voice Calls	₹66,400
AWS Elastic Beanstalk (t3.medium)	₹25,000
AWS S3 (Storage + Hosting)	₹2,500
AWS Bedrock (Nova Lite)	₹8,300
Total (approx)	₹120,460 / month

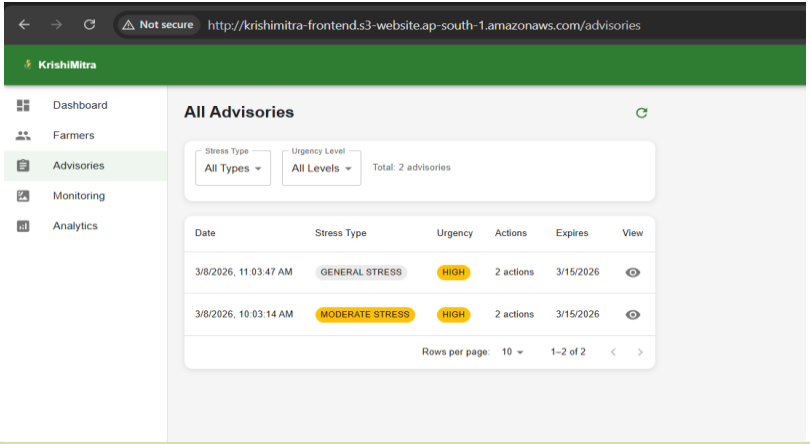
💡 Optimized Cost

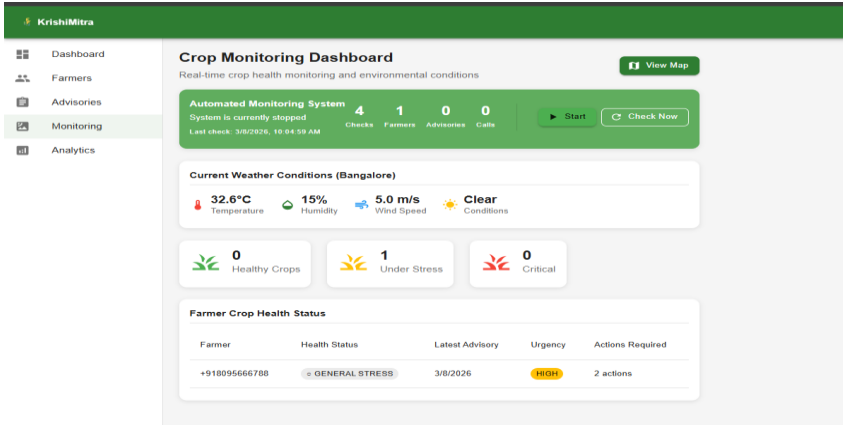
Plan	INR
Optimized possible cost	**₹60,000 / month**
Per farmer cost	**₹6–₹12 / month** (approx)

Snapshots of the prototype:









Prototype Performance report/Benchmarking

Performance highlights

- Crop disease detection accuracy: **~90–95%**
- Response latency: **< 3 seconds**
- AI query processing: **2–4 seconds**
- Image processing time: **1–2 seconds**

System scalability

- Cloud deployment supports concurrent users
- Microservice architecture allows independent scaling.

Additional Details/Future Development

Future enhancements

- Voice-based farmer interaction
- Regional language support for all Indian languages
- Personalized crop recommendations
- Soil health analysis integration
- AI-based yield prediction
- Mobile app version for Android devices

Goal

Make KrishiMitra a complete digital assistant for Indian farmers.

Prototype Assets:

- ➔ GitHub Public Repository Link : Krishimitra
- ➔ Demo Video Link: Demo Video

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Thank You